

A COMPLETE PRACTICAL GUIDE TO MASTER
LARGE LANGUAGE MODELS, RAG, AGENTS & GENERATIVE AI



UPDATED & EXPANDED

COVERS THE LATEST
ADVANCEMENTS &
ENTERPRISE PRACTICES

LLM ENGINEERING & RAG MASTERY

— ZERO TO HERO —



PROMPT
ENGINEERING

RAG

AGENTS

DEPLOYMENT
& MLOPS

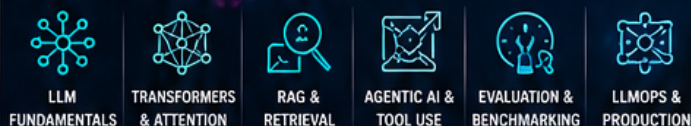
VECTOR
DATABASES

SAFETY &
ALIGNMENT

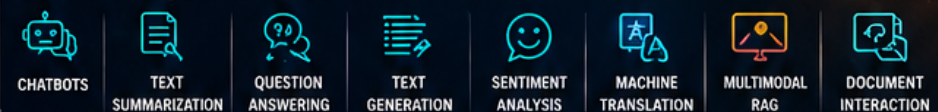
BUILD, FINETUNE, AND DEPLOY PRODUCTION-READY
AI SYSTEMS WITH LLMs, RAG & AGENTS

- End-to-End Coverage: LLMs, Prompting, Fine-Tuning, RAG, Agents & LLMOps
- Master Transformers, Tokenization, Embeddings & Model Internals
- Build RAG Systems with Vector DBs, Retrieval, Chunking & Advanced Techniques
- Fine-Tune Models with LoRA, QLoRA, RLHF, PEFT, Distillation & More
- Deploy, Optimize & Monitor LLMs in Production at Scale
- Real-World Projects, Best Practices & Interview Preparation

MASTER CORE CAPABILITIES



SOLVE REAL-WORLD TASKS



POWERED BY LEADING TOOLS & FRAMEWORKS



BY ABHISHEK PANDA

AI ARCHITECT | AZURE AI ARCHITECT



BEGINNER TO
ADVANCED



PRACTICAL.
REAL. IMPACTFUL.



INDUSTRY-READY
SKILLS



FUTURE-PROOF
YOUR CAREER



LLM ENGINEERING & RAG MASTERY: **FROM ZERO TO HERO**

BY ABHISHEK PANDA
AI ARCHITECT | AZURE AI ARCHITECT

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- 1.2 EVOLUTION OF LANGUAGE MODELS
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- 1.7 LLM LIFECYCLE OVERVIEW

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UNDERSTANDING INTERNALS, TRAINING, ADAPTATION

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- PYTHON
- JUPYTER / VS CODE
- PYTORCH
- TENSORFLOW (OPTIONAL/LIGHT COVERAGE)
- HUGGING FACE TRANSFORMERS
- DATASETS
- TOKENIZERS
- ACCELERATE
- PEFT
- TRL
- BITSANDBYTES

MODELS USED

OPEN SOURCE

- LLAMA FAMILY
- MISTRAL / MIXTRAL
- FALCON
- PHI MODELS
- GEMMA
- DEEPSEEK / QWEN / YI (IF RELEVANT BY PUBLICATION TIME)
- KIMI
- SEEDDANCE

CLOSED SOURCE

- OPENAI GPT MODELS
- ANTHROPIC CLAUDE
- GOOGLE GEMINI

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- PROMPT PLAYGROUNDS / NOTEBOOKS
- OPENAI PLAYGROUND / AZURE OPENAI STUDIO
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- LANGSMITH (EVALUATION/TRACE OPTIONAL)

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TOOLS & LIBRARIES

CORE RAG FRAMEWORKS

- **LLAMAINDEX**
- **LANGCHAIN RETRIEVAL MODULES**
- **HAYSTACK (OPTIONAL ADVANCED ALT)**

EMBEDDING MODELS

OPEN SOURCE

- **BGE**
- **E5**
- **INSTRUCTOR**
- **SENTENCE TRANSFORMERS**
- **MINILM VARIANTS**

CLOSED SOURCE

- OPENAI EMBEDDINGS
- AZURE OPENAI EMBEDDINGS
- COHERE EMBEDDINGS (OPTIONAL)

VECTOR DATABASES

LOCAL DEVELOPMENT FIRST

- FAISS
- CHROMADB
- LANCEDB

PRODUCTION / CLOUD / ENTERPRISE

- PINECONE
- WEAVIATE
- QDRANT
- MILVUS
- AZURE AI SEARCH

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TOOLS & LIBRARIES

AGENT FRAMEWORKS

- **LANGGRAPH** (RECOMMENDED FLAGSHIP)
- **CREWAI**
- **AUTOGEN**
- **SEMANTIC KERNEL**
- **PYDANTICAI** (MODERN EMERGING OPTION)

LOCAL AGENT DEVELOPMENT

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TOOLS & LIBRARIES

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TOOLS & LIBRARIES

SERVING / OPTIMIZATION

- **VLLM**
- **TGI (TEXT GENERATION INFERENCE)**
- **TENSORRT-LLM**
- **OLLAMA (LOCAL SERVING/DEV)**
- **LLAMA.CPP (EDGE/LOCAL INFERENCE)**

CONTAINERIZATION / INFRA

- **DOCKER**
- **DOCKER COMPOSE**
- **KUBERNETES**
- **HELM**

MONITORING / OBSERVABILITY

- **LANGSMITH**
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- **PHOENIX ARIZE**
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- **WEIGHTS & BIASES**

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AZURE STACK

- **AZURE OPENAI**
- **AZURE AI SEARCH**
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APPENDIX E — AZURE AI CERTIFICATION ROADMAP

RECOMMENDED TOOLS / FRAMEWORKS COVERED

1. HUGGING FACE TRANSFORMERS
2. PEFT
3. TRL
4. LANGCHAIN
5. LANGGRAPH
6. LANGSMITH
7. CREWAI
8. SEMANTIC KERNEL
9. LLAMAINDEX
10. FAISS
11. CHROMA
12. PINECONE / WEAVIATE / QDRANT
13. VLLM
14. TENSORRT-LLM
15. MLFLOW
16. WEIGHTS & BIASES
17. AZURE OPENAI
18. AZURE AI SEARCH
19. AZURE AI FOUNDRY
20. PROMPT FLOW

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